

Measuring Human Behaviors and Psychological Constructs for ADHD: A Review on emerging techniques, and their challenges

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Abstract—Attention Deficit Hyperactivity Disorder (ADHD) is difficult to diagnose, along with many other mental disorders. The diagnosis process for ADHD in Ireland usually involves an individual's history, symptoms throughout their life, and assessments in the form of psychological or medical evaluations. At the very least, a clinical interview should be conducted by a specialist, who may be flexible and adaptable in their approach and procedure, as ADHD can appear differently between people [7] [1]. Measuring one or more of the core traits of ADHD, which are Inattention, Hyperactivity and Impulsivity, could lead to an easier and a quicker form of diagnosis. Therefore, this review paper will discuss techniques and implementations from recent literature for measuring one or more of these traits.

Index Terms—Accessibility, Video Games, Human Computer Interaction, Brain Computer interfaces, Electroencephalogram, Electroencephalography

I. INTRODUCTION

In both clinical assessment and neuroscience research, a diagnosis of Attention Deficit Hyperactivity Disorder (ADHD) can be complicated. A proper clinical diagnosis usually involves multiple assessments: one or more clinical interviews, questionnaires, and Continuous Performance Tests (CPT) [1] [7]. Recent research has shown that emerging techniques can be used to measure one or more of the core traits of ADHD, which are Inattention, Hyperactivity, and Impulsivity [13]. Furthermore, if these techniques are trustworthy, then they could potentially be used to provide a quicker, more simple form of diagnosis for ADHD.

Studies in the literature have identified several techniques that can be used to provide a better diagnosis of ADHD, including but not limited to: Brain Computer Interfaces (BCI), Self-Report tools with Behavioral tasks, and Deep Inverse Reinforcement Learning techniques [11] [4] [20]. These emerging techniques are rarely used or compared against in most studies regarding ADHD due to their novel approach, or their complicated setup. In contrast to these novel approaches, the standard clinical and research behavioral tasks, such as CPTs for ADHD include approaches like the commonly used Test Of Variables Of Attention (TOVA), the Gordon Diagnostic System (GDS) or the QbTest [8]. Overall, while these techniques

individually have weaknesses, they can be combined together in order to provide a much more accurate diagnosis of ADHD, and potentially other mental disorders.

Self-Report tools have been effective for any and all diagnosis, with the exception of complicated conditions, such as mental disorders, as Self-Report tools require the user to be completely honest and unbiased when providing their answers. In contrast, Behavioral tasks have been more effective for diagnosing conditions, as their diagnosis relies on the results of the task. However, in previous research, the gap between Self-Report and Behavioral tasks has invalidated the results of many different research papers. Furthermore, this gap is a problem for diagnosing the core traits of ADHD, as both Self-Report, and Behavioral tasks are both used and relied upon for the diagnosis.

There has been interest in Deep Inverse Reinforcement Learning recently, as modern and more sophisticated Inverse Reinforcement Learning Algorithms have demonstrated the ability to Model psychological constructs, such as the core traits of ADHD. Similarly, recent studies have shown that Deep Inverse Reinforcement Learning can be used to measure and predict Human Behavior.

The aim of this review paper is to provide a way forward through a critical analysis of these emerging tools and their implementations only. Another aim of this review is to highlight multiple research papers that employ these various techniques. Due to many papers which already cover different types of CPTs and their performance, CPTs will only be mentioned in the context of behavioral tasks.

II. APPROACHES FOR DIAGNOSING ADHD

A. Brain-Computer Interfaces

Brain-Computer Interfaces are systems that allow brain-computer interactions by recording, processing and taking action with data from the brain. This is usually done by using recording devices like EEGs in conjunction with AI-algorithms to perform an action like moving a mouse, or typing on a computer [9].

Herbert (2024) [9] mentions that while P300 BCIs and sensorimotor (SMR) BCIs work by using, or relying on attention, human behavior plays a role in the performance of these BCIs [9]. The main point made in this paper is that positive mood and self-views can broaden attention, thereby raising the performance of BCIs, while negative mood and self-views can lower the performance of BCIs. One of the conclusions made relates to self-report, and how the BCI performance of the first recording as a baseline can be used with self-report in order to raise BCI performance. This is done by having the user complete a self-report after the baseline, and then positively motivating or encouraging the user as necessary in order to positively affect their mood, thereby raising the BCI performance.

However, Mashrur (2022) [16] disproves some of the validity of the previous paper by stating that self-report tools are frequently criticized for "failing to extract true consumer preferences" [16]. In its place, a new framework is introduced, which classifies consumer choice from EEG signals. The classification methods that were tested include Naive Bayes, Decision Trees, Support Vector Machines and K-Nearest Neighbors, which contrasts with other mentioned papers in later sections of this review, specifically Section 3, which goes over the use of Deep Inverse Reinforcement Learning models for extracting human behavior.

Similarly, through the research conducted by Carelli (2017) [3] and her colleagues in their paper, they agree with Herbert (2024) by mentioning that the performance of P300 BCIs is not solely based on attention, but that other factors such as general intelligence and working memory also play a role in the performance of the BCI systems. They mention that working memory training can improve BCI usability and performance. This aligns with Herbert's (2024) [9] statement about effectively using other factors like mood in order to encourage users positively after their self-report, which allows for potentially better BCI performance. There is also a mention of another paper from Rohani (2015) [19] in Carelli's (2017) [3] paper which uses a P300 BCI and VR in order to train attention abilities in ADHD patients, further supporting the point that P300 BCIs can be used for attention. Overall, the main conclusion of this paper is that P300 BCIs can be used for the diagnosis of mental disorders like anxiety and ADHD by measuring attention, and that many papers have already implemented and tested systems that can help ADHD patients.

Carelli (2017) [3] also highlights the importance of Ecological Validity. They state that "the realization of engaging settings for cognitive training, sometimes involving gaming systems, entails the realization of more realistic and interactive protocols that needs to be consistently adopted across several studies for standardization" [3]. They are stressing the importance of realistic research, as lab settings and real-life situations are often unrelated. This Ecological Validity is mentioned in depth in the 'Bridging the Gap...' paper by Lee (2024) [11] and their team, and it is also one of the limitations of their experiment.

While BCI projects that measure hyperactivity are scarce,

one research paper explains that the first attempt to diagnose ADHD using BCIs involved studying brain signals of children with behavior problems such as hyperactivity. These brain signals showed slow waves (Gonzalez et al, 2022, as cited in Jasper, 1938, and Jouzizadeh, 2020) [15]. This paper does go on to mention that inattention has been correlated with delta band changes, while hyperactivity and impulsivity have been correlated with the delta-theta ratio (Gonzalez et al, 2022, as cited in Bashiri, 2018) [15]. This claim is supported by another paper, which states that an increase in sensation seeking or hyperactivity could be related to the fast decline of EEG vigilance during rest [18].

Overall Herbert's (2024) [9] paper reinforces the idea of using BCIs for dealing with attention (or inattention in the case of ADHD) related research. It also mentions human behavior in the form of mood and self-views, which can affect the performance of BCIs due to increases or decreases in attention. Self-Report is used in this paper to capture human behavior by assessing the mood or self-views of the user taking the Self-Report [9]. Mashrur (2022) [16] conflicts with Herbert's paper by stating that while questionnaires or self-report tools are simple, they "oftentimes fail to reflect the true state of mind" of people [16]. So we can conclude that using self-report to measure human behavior is ambiguous, and that exploring other techniques (such as Deep Inverse Reinforcement Learning) could yield better results when measuring human behavior. One point that stands out between these papers is that human behavior is not defined clearly. Herbert (2024) uses mood and self-views as examples of human behavior, while Mashrur (2022) implies that human behavior is preferences. While Mashrur's paper is older, its definition of human behavior fits inverse reinforcement learning, where preferences could be modeled as rewards. Carelli (2017) [3] adds onto the literature by stating that other factors like general intelligence and working memory play important roles in both the performance and usability of BCI systems. From these papers, we can assume that BCIs are very effective for measuring inattention in ADHD users, and that human behavior could potentially affect BCIs, although the definition of human behavior is not standardized. Hyperactivity is rarely mentioned, however hyperactivity could also be one of the more easier traits to diagnose, along with impulsivity. One online article by Tom Murfit (2024) [14] supports this hypothesis, stating that people who have ADHD without hyperactivity, better known as ADHD Predominantly Inattentive Type (ADHD-PI), usually behave in a more quiet and reserved way, and are not disruptive. This article makes a comparison to regular ADHD, which is stated to be more obvious and easier to diagnose, most likely due to focusing on hyperactive signs.

B. Self-Report Tools and Behavioral Tasks

Both Self-Report Tools and Behavioral Tasks are widely used for diagnosing ADHD traits. Behavioral assessments, such as a clinical interview, are helpful for assessing hyperactivity. In these interviews, the observee who possibly has ADHD, is watched with the goal of assessing one or

more core traits of ADHD. While the previous section of this review concludes with evidence towards hyperactivity being easy to assess, ADHD-PI may not be as obvious. There are several papers which go over the use of Self-Report, Behavioral Tasks, or both for assessing human behaviors. However, there is usually a discrepancy between Self-Report Tools and Behavioral Tasks.

'Bridging the Gap...' by Lee et al. (2024) is a paper about connecting both Self-Report and Behavioral Tasks together through Deep Inverse Reinforcement Learning [11]. This paper explains the problem of Self-Report and Behavioral Tasks not lining up. Although both methods are commonly used individually or together, their validity is often questioned. On one hand, it is difficult to know if Self-Report responses are biased or not, as many factors such as social desirability, a consistent response, or always extreme / always midpoint answers affect the responses. As mentioned in the Brain Computer Interfaces section, Mashrur (2022) [16] raises the point that Self-Report is often criticized for its inability to capture true human preferences. On the other hand, Lee et al. (2024) states that behavioral tasks work well due to users engaging in the task in a mostly non-biased way. However, the problem with behavioral tasks arises when applying simple statistics or descriptive analysis to the data from the behavioral task. Lee et al. (2024) offers a solution to these problems by opting to use Inverse Reinforcement Learning instead of simple statistics for data from behavioral tasks. The output from the IRL model is then compared to the Self-Report tool known as the Baratt Impulsiveness Scale (BIS). This scale is used to measure impulsivity in users by asking a series of questions, and allowing users to respond via a Likert Scale from Always to Never. The conclusion of this paper ends with Lee et al. (2024) proposing that behavioral real-time tasks will only reflect self-reported traits when the analysis method used on the tasks data reflects the complexity of the task [11].

This solution of using a complicated analysis method like IRL for behavioral tasks is useful, as Demidenko et al. (2019) directly brings up the problem of self-report and behavioral measures not lining up [6]. They state that self-report and behavioral tasks have not received enough attention when it relates to predictive validity and contrasting results. A common concern amongst these papers so far have been mostly questioning if self-report or behavioral tasks can be trusted. It appears that the most difficult part of diagnosing ADHD is measuring impulsivity. Demidenko et al. (2019) is a paper that complements the paper by Lee et al. (2024) [11] well. Demidenko et al. (2019) does not use the word human behavior, and instead uses the term "risk behavior". They mention that risk behavior is broken up into two parts, which are reward sensitivity, and self-regulation. This standardization of behavior provides a solution to defining human behavior in the form of something tangible to measure (risk behavior), which is the reward sensitivity, and self-regulation of a user. They define reward sensitivity as the motivation to do something, and self-regulation is defined as a certain behavior that is unable to be inhibited. Previously mentioned

papers that mention human behavior, such as Mashrur (2022) [16], Herbert (2024) [9], and Lee et al. [11] should be looked at again, to see if either the researchers were actually studying risk behavior, or if there is another sort of behavior that was being assessed. Finally, Demidenko et al. (2019) defines risk behavior as acting without forethought, which sounds similar to the used definition of impulsivity (impulsive action, impulsive choice, and impulsive personality) in the paper by Lee et al. (2024) [11].

One older paper that supports Lee et al. (2024) [11] in both their research and implementation of self-report and behavioral tasks is from Moskowitz (1986) [17]. In this paper, Moskowitz (1986) claims that self-report and behavioral tasks should be evaluated by Loevinger's concept of structural validity. Loevinger believed that the ego is a process. Therefore, Moskowitz (1986) called for all future papers to have scores of behaviors or characteristics that could be generalized, and that paralleled the definition of the behavior that was being mentioned. This was already seen in the paper by Lee et al. (2024), where they gave a definition for impulsivity, and used Inverse Reinforcement Learning, which is a model that can be generalized, in order to output a score for impulsivity. In contrast to Inverse Reinforcement Learning, Simple statistics can not be generalized, as there may be different data processing techniques between researchers. Simple statistics also cannot be used as something complicated like impulsivity cannot be represented with statistics. Moskowitz (1986) also states that the process for finding convergence is often misguided. Convergence in this paper means bridging the gap between self-report and behavioral tasks. Moskowitz (1986) recommends that while convergence is difficult, if convergence is ever found, it needs to be for one aspect of a behavior and characteristic, and not the whole behavior and characteristic. Lee et al. (2024) follows this rule well, as they are not trying to predict ADHD, but rather an aspect of ADHD, which is impulsivity. Another similarity between the two papers is that Moskowitz (1986) mentions that behavioral tasks can provide valid assessments of characteristics. Lee et al. (2024) also adheres to this rule by using the data from the real time behavioral task only to measure impulsivity. The self-report tool is then used to complement the task, which is how they bridge the gap between the behavioral task and the self-report tool.

One conclusion that we can make after comparing these papers is that both self-report tools and behavioral tasks are efficient at measuring different human behaviors or characteristic. However, achieving "convergence", as mentioned in the paper by Moskowitz (1986) [17] is often difficult, and that researchers should not be surprised if they can not find convergence between behavioral tasks and self-report tools. Demidenko et al. (2019) [6] agrees with Moskowitz, by stating that not enough focus is placed on the validity and contrasting results of self-report and behavioral tasks. One issue with a lot of papers that deal with human behaviors, or psychological constructs is that they often find their own definition of human behavior, or they vaguely use the word human behavior to

describe constructs. Demidenko et al. (2019) proposes the term 'risk behavior', which is broken up into reward sensitivity and self-regulation. Moskowitz (1986) mentions Loevinger, who created 9 stages for ego development. reward sensitivity and self-regulation are implicitly seen in Loevingers 9 stages, either in lower stages (reward-sensitivity), or as an output of the 9th stage (self-regulation). We can effectively conclude that Demidenko et al. (2019) has generally kept up with, and built upon the literature, by further developing a behavior framework through 'risk behavior'. Demidenko et al. (2019) has also specifically defined what risk behavior means, and the factors that drive it (reward-sensitivity and self-regulation). Lee et al. (2024) [11] has successfully performed research that takes into account concerns found by Moskowitz (1986) and Demidenko et al. (2019) on self-reports and behavioral tasks. However, Lee et al. (2024) does not explain what human behavior is, or what it consists of. In contrast to this, Lee et al. does state that impulsivity is defined differently in a wide variety of papers, and that it is unclear what construct of impulsivity is being measured, or what construct is correct. Lee et al. (2024) implements the idea of convergence by Moskowitz (1986) with an explanation of what impulsivity is, and what aspect of impulsivity they measured (impulsive action, impulsive choice, and impulsive personality). They then conduct their research by measuring this type of impulsivity using an IRL model. Therefore, future research that includes both self-report and behavioral tasks should first define the behavior they are measuring, and what characteristics of the behavior is taken into account.

C. Inverse Reinforcement Learning

While BCI systems are affected by human behavior, and self-report / behavioral tasks rely on a clear definition of human behavior and its characteristics, Inverse Reinforcement Learning is stated to be able to predict human behavior by inferring the underlying reward and preferences of humans [10]. Ettinger (2019) presents the idea of the 'Theory of Mind', which is our ability to predict other people's mental states by using hypothesized beliefs and desires of that person. Ettinger (2019) states that the Theory of Mind is no different to IRL, which infers an agents reward function given their actions.

Das (2014) uses IRL in order to predict human behavior, and how it changes in response to social media feedback [5]. However, one issue that is present with this research paper is that it does not define human behavior. Previously, in the BCI section, Demidenko et al. (2019) mentioned that one common problem of research papers that mention human behavior, is that they do not define what it is, or what characteristics are used. Das (2014) does not mention any psychological constructs, or any characteristics either. As a result, this research paper is harder to interpret. Figure 1 of this paper shows the "Behavior changes by reply count", but we don't know what behavior has changed.

IRL is also used to predict risky decision making in Liu et al.'s (2019) paper [12]. In the self-report and behavioral tasks section, Demidenko et al. (2019) mentions the concept

of risk behavior [6]. Liu et al. (2019) does not define human behavior as well but it could be implied that Liu et al. (2019) is predicting risk behavior. Liu et al. (2019) also uses an IRL model that scores higher on risk-prone groups, compared to risk-averse groups, suggesting that IRL is better at predicting risk, or perhaps even impulsivity for ADHD.

The ideas gathered from these papers propose that Inverse Reinforcement Learning is able to model behaviors from humans, and that IRL is best suited for predicting risk-prone behavior [12]. This is a promising technique for predicting impulsivity in users who have ADHD or ADHD-PI. However, these papers suffer from the same problems that some of the other papers on self-report, behavioral tasks or BCI have. They do not define behavior clearly, so it is implied that human behavior includes all behaviors. One way to possibly improve these research papers would be to clearly define the characteristics of either the behavior, or the environment, and to state what characteristics are likely to occur. An example of this problem is in Das's (2014) paper, where addiction or risk is not mentioned in the entire paper, which are both likely to occur with social media use. More information on social media addiction, human behaviors on social media, and its risks can be found in Amirthalingam's (2024) paper[2].

III. DISCUSSION

In this review, we went over emerging techniques, as well as their drawbacks, when predicting human behavior. Behavior needs to be either more specific, or broken up into characteristics that can be measured. We see an example of this in Demidenko et al.'s (2019) [6] paper, where Demidenko et al. criticizes the ambiguity of the term human behavior, and proposes risk behavior, which is further broken down into two characteristics (reward sensitivity and self-regulation).

ADHD's 3 core traits can be predicted or measured using a wide variety of techniques such as BCI systems, self-report tools, behavioral tasks, and Inverse Reinforcement Learning models. BCIs are capable at measuring attention through the use of P300 BCIs for example [9] [16] [3] These BCI systems work by using EEGs in order to record electrical voltage changes in the brain to produce an Event Related Potential. Since these ERPs occur due to auditory or visual stimuli, they are useful for measuring inattention, a core trait of ADHD. While hyperactivity is rarely mentioned, most likely due to its ease of detection by clinical observations [14], BCI systems can also measure hyperactivity by looking for slow waves in brain signals [15] [18]. The drawbacks of BCI systems are that they can be influenced by human behavior. Positive mood or self views can broaden attention, while negative mood or self views may hinder attention [9].

Self-report tools and behavioral tasks are useful for measuring behaviors and characteristics. The diagnosis for ADHD often involves a combination of both self-report tools and behavioral tasks in the form of behavioral assessments and questionnaires. Behavioral assessments such as clinical observations are also one of the main ways to measure hyperactivity and impulsivity. However, self-report and behavioral tasks

usually aren't related to each other [11] [17] [6]. Behavioral tasks can provide valid assessments of characteristics, but Moskowitz (1986) mentions that the characteristic or behavior should be clearly defined. If the characteristic or behavior is clearly defined, then convergence between self-report and behavioral tasks is more likely to occur. Self-report tools are usually used as a complement or a confirmation of behavioral assessments. While self-report tools are helpful, there can be a lot of bias from self-report tools, as users may not answer honestly.

Inverse Reinforcement Learning can replace simple statistics for predicting behaviors and characteristics. This is due to IRL models being able to learn the reward functions of agents or humans [10]. IRL models are especially powerful when used alongside data from real time behavioral tasks [11]. IRL models suffer from the same problem as previously mentioned papers, as the behavior or characteristic needs to be clearly defined in order to interpret the reward function produced by the IRL model. Overall, these emerging techniques work well together, and provide an accurate prediction or measure of human behavior when combined into one project, as seen in the research conducted by Lee et al.

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